

Unit 5.5Indefinite IntegralsDefinition :

$\int f(x) dx$ is called an indefinite integral and

$\int f(x) dx = F(x) + C$ if and only if

$$F'(x) = f(x).$$

Properties :

$$(i) \int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$$

$$(ii) \int [f(x) - g(x)] dx = \int f(x) dx - \int g(x) dx$$

$$(iii) \int C f(x) dx = C \int f(x) dx$$

STANDARD FORMULAE

$$\int k dx = kx + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$\int e^x dx = e^x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \frac{1}{x^2+1} dx = \tan^{-1} x + C$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

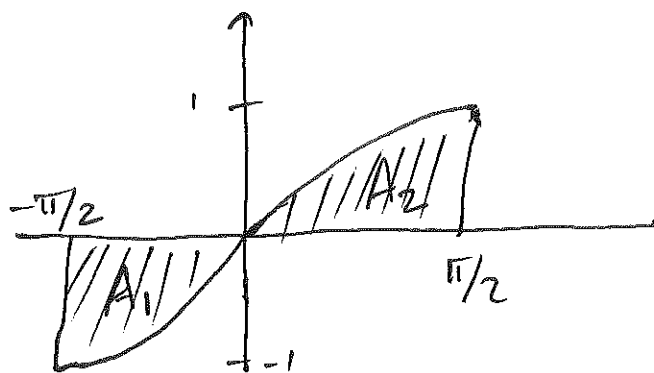
Example 1

a) Find $\int \sin x \, dx$ Solution
 $= -\cos x + C$

b) Evaluate $\int_{-\pi/2}^{\pi/2} \sin x \, dx$ Sol
 $= -\cos x \Big|_{-\pi/2}^{\pi/2}$
 $= -\left[\cos \frac{\pi}{2} - \cos -\frac{\pi}{2}\right]$
 $= -[0 - 0]$
 $= 0$

c) Find the area between the graph of $f(x) = \sin x$, the x -axis, the lines $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{2}$

Solution: Always sketch if the area is involved.



Area of the enclosed region is

$$A = A_1 + A_2$$

$$= -\int_{-\pi/2}^0 \sin x \, dx + \int_0^{\pi/2} \sin x \, dx$$

$$= -\left[-\cos x \Big|_{-\pi/2}^0\right] + \left[-\cos x \Big|_0^{\pi/2}\right]$$

$$= -\left[-\cos 0 - -\cos -\frac{\pi}{2}\right] + \left[-\cos \frac{\pi}{2} - -\cos 0\right] = -[-1 + 0] + [0 + 1] = 2$$

(3)

Examples: Evaluate $F(x)$ where $\frac{d}{dx} F(x) = f$, the integrand

$$1) \int (1-2x)^5 dx = F(x) + C$$

$$\Rightarrow \int (1-2x)^5 dx = \frac{1}{-12} (1-2x)^6 + C$$

Guess:

$$F(x) = (1-2x)^6 \text{ But}$$

$$\frac{d}{dx} F(x) = 6(1-2x)^5 \cdot -2 = -12(1-2x)^5$$

We do not need -12 for the integrand f .

Thus

$$2) \int \sin^5(2\theta) \cos(2\theta) d\theta = F(x) + C$$

$$\Rightarrow \int \sin^5(2\theta) \cos(2\theta) d\theta = \frac{1}{12} \sin^6(2\theta) + C$$

Guess:

$$F(x) = (\sin^6 2\theta) \text{ But}$$

$$\frac{d}{dx} F(x) = 6 \sin^5(2\theta) \cdot \cos(2\theta) \cdot 2 = 12 \sin^5(2\theta) \cos(2\theta)$$

everything is OK for the integrand f except 12.

Thus

$$3) \int e^{3t} dt = F(x) + C$$

$$\int e^{3t} dx = \frac{1}{3} e^{3t} + C$$

Guess:

$$F(x) = e^{3t} \text{ But}$$

$$\frac{d}{dx} F(x) = e^{3t} \cdot 3 = 3e^{3t} \text{ We}$$

do not have 3 in the integrand f .

Thus

$$4) \int \frac{\sin 3x}{1 - \cos 3x} dx = F(x) + C$$

$$\int \frac{\sin 3x}{1 - \cos 3x} dx = \frac{1}{3} \ln |1 - \cos 3x| + C$$

$$\text{Guess: } F(x) = \ln |1 - \cos 3x|$$

$$\text{But } \frac{d}{dx} F(x) = \frac{1}{1 - \cos 3x} \cdot \sin 3x \cdot 3$$

We do not need 3 in the integrand f . Thus

5) $\int \frac{x}{9+x^4} dx = F(x) + C$

Rewrite first

$\Rightarrow \int \frac{x}{9[1+(\frac{x^2}{3})^2]} dx = F(x) + C$

$= \frac{1}{9} \int \frac{x}{1+(\frac{x^2}{3})^2} dx = F(x) + C$

$\frac{1}{9} \int \frac{x}{1+(\frac{x^2}{3})^2} dx = \frac{1}{6} \tan^{-1}(\frac{x^2}{3}) + C$

Now Guess:
 $F(x) = \tan^{-1}(\frac{x^2}{3})$

But

$\frac{d}{dx} [\tan^{-1}(\frac{x^2}{3})]$

$= \frac{1}{1+(\frac{x^2}{3})^2} \cdot \frac{2}{3} x$

$= \frac{2}{3} \frac{x}{1+(\frac{x^2}{3})^2}$

We need $\frac{1}{9} \frac{x}{1+(\frac{x^2}{3})^2}$
 Thus

6. $\int \frac{u}{1+u^2} du = F(x) + C$

$= \frac{1}{2} \ln |1+u^2| + C$

$\tan^{-1} u + C$
 wont work

check by
 $\frac{d}{dx} (\frac{1}{2} \ln |1+u^2|)$

7. $\int \frac{3}{(1-4x)^2} dx = F(x) + C$ (Rewrite)

$\int \frac{3}{(1-4x)^2} dx = 3 \int (1-4x)^{-2} dx = F(x) + C$

$\int \frac{3}{(1-4x)^2} dx = \frac{3}{4} (1-4x)^{-1} + C$

Guess!

$F(x) = (1-4x)^{-1}$

But

$\frac{d}{dx} (1-4x)^{-1} = (1-4x)^{-2} \cdot -4$
 $= -4(1-4x)^{-2}$

we need $3(1-4x)^{-2}$

Thus

$$8) \int \frac{1}{1-4x} dx = \frac{-1}{4} \ln|1-4x| + C$$

$$\begin{aligned} 9) \int \frac{dx}{\sqrt{1-7x}} &= \int (1-7x)^{-\frac{1}{2}} dx \\ &= F(x) + C \\ &= (1-7x)^{\frac{1}{2}} \cdot 2 \cdot \frac{1}{-7} + C \\ &= -\frac{2}{7} (1-7x)^{\frac{1}{2}} + C \\ &= -\frac{2}{7} \sqrt{1-7x} + C \end{aligned}$$

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Since

$$\begin{aligned} \frac{d}{dx} \left(\frac{-1}{4} \ln|1-4x| + C \right) &= \frac{-1}{4} \cdot \frac{1}{1-4x} \cdot -4 \\ &= \frac{1}{1-4x} = f(x), \end{aligned}$$

the integrand.

10) Sometimes we have to rewrite the function

$$\begin{aligned} \int (1-x^3)^2 dx &= \int (1-2x^3+x^6) dx \\ &= \int dx - 2 \int x^3 dx + \int x^6 dx \\ &= x - \frac{2x^4}{4} + \frac{x^7}{7} + C \end{aligned}$$

$$\begin{aligned} 11) \int \frac{1+x^5}{x^4} dx &= \int \frac{1}{x^4} dx + \int \frac{x^5}{x^4} dx \\ &= \int x^{-4} dx + \int x dx \\ &= -\frac{1}{3} x^{-3} + \frac{1}{2} x^2 + C \end{aligned}$$

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$$12) \int \frac{x^4}{1+x^{10}} dx = \int \frac{x^4}{1+(x^5)^2} dx$$

$$= \frac{1}{5} \tan^{-1}(x^5) + C$$

Guess!
 $F(x) = \tan^{-1} x^5$. But
 Since $\frac{d}{dx} F(x) = \frac{1 \cdot 5x^4}{1+(x^5)^2}$
 $F(x)$ has to be.

$$13) \int \frac{x^2}{\sqrt{16-x^6}} dx$$

Rewrite.
 thinking of
 an arcsin.

$$= \int \frac{x^2}{\sqrt{16(1 - \frac{x^6}{16})}} dx$$

$$= \int \frac{x^2}{\sqrt{16(1 - (\frac{x^3}{4})^2)}} dx$$

$$= \int \frac{1}{4} \frac{x^2 dx}{\sqrt{1 - (\frac{x^3}{4})^2}}$$

✗

$$= F(x) + C$$

$$= \frac{1}{3} \sin^{-1}\left(\frac{x^3}{4}\right) + C$$

Guess
 $F(x) = \sin^{-1}\left(\frac{x^3}{4}\right)$
 But since
 $\frac{d}{dx} \sin^{-1}\left(\frac{x^3}{4}\right) = \frac{1}{\sqrt{1 - (\frac{x^3}{4})^2}} \cdot \frac{3x^2}{4}$
 $= \frac{1}{4} \frac{3x^2}{\sqrt{1 - (\frac{x^3}{4})^2}}$
 $F(x)$ has to be
 $= \frac{1}{3} \sin^{-1}\left(\frac{x^3}{4}\right)$

Note $\int \frac{x^5}{\sqrt{16-x^6}} dx$

Rewrite
 thinking of
 powers

$$= \int x^5 (16-x^6)^{-1/2} dx = F(x) + C$$

$$\Rightarrow \int x^5 (16-x^6)^{-1/2} dx = -\frac{1}{3} \sqrt{16-x^6}$$

Guess.
 $F(x) = (16-x^6)^{1/2}$
 But since
 $\frac{d}{dx} F(x) = \frac{1}{2} (16-x^6)^{-1/2} \cdot (-6x^5)$
 $= -3 \frac{x^5}{\sqrt{16-x^6}}$
 everything ok except
 -3

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$$14) \int (x^2 e^{x^3} \sec e^{x^3} \cdot \tan e^{x^3}) dx = f(x) + C$$

$$= \frac{1}{3} \sec e^{x^3} + C$$

Guess:

$$f(x) = \sec(e^{x^3})$$

and modify it

$$15) \int \frac{\sec^2 x}{\sqrt{4 + \tan x}} dx = \int \sec^2 x (4 + \tan x)^{-1/2} dx$$

$$= 2 (4 + \tan x)^{1/2} + C$$

Guess
 $(4 + \tan x)^{1/2}$
 but corrected
 the $\frac{1}{2}$ by
 2.

H/W Evaluate

$$16) \int \frac{\operatorname{cosec}^2(x)}{x} dx$$

$$17) \int \frac{\sin(\frac{1}{x})}{x^2} dx$$

$$18) \int 8^{\operatorname{cosec}(5x)} \operatorname{cosec}(5x) \cot 5(x) dx$$

$$19) \int x \cosh(x^2) 3^{\sinh(x^2)} dx$$